# Bulgaria: energy security, economic opportunity and partnership priorities

POLICY MEMO · COUNTRY INSIGHTS · ALLIED INDUSTRIAL BASE EXPLORER · NEIS CENTER

*AI-generated from the data and checked against it; paper baseline as of 12 October 2026.*


## Summary

- Bulgaria is most exposed in upstream gas, downstream wind and upstream electric vehicles, with Energy Security higher than only 10%, 16% and 29% of the other economies scored. In upstream gas, unrecorded reserves and a trade deficit pull it down.[1]
- Its strongest openings are in downstream batteries, upstream electric grid and downstream solar, with Economic Opportunity higher than 96%, 92% and 88% of the other economies scored; investment or trade lifts each.[1]
- On the paper's weighting, its top Strategic Index sectors are upstream batteries, midstream batteries and upstream solar, almost tied. In upstream batteries, midstream batteries, midstream electric grid and midstream solar, top priority meets the most policy.[1]
- The Partnership Strength Index rates Italy highest, then Germany. They are also its 3rd and 1st trade partners. China and Turkiye are each a strong tie today that the index ranks lower.[1,2]


## 1. Energy security risks and why

10 of Bulgaria's 29 placed sectors, those whose scores can be compared with other economies, sit in the two low-energy-security corners. Energy Security measures how well a country can rely on what it needs in a sector. Compared with other economies, its Energy Security is lowest in upstream gas, downstream wind, upstream electric vehicles and upstream batteries. Its scores there are higher than only 10%, 16%, 29% and 32% of the other 97, 92, 93 and 76 economies scored, and all sit in the Secure corner.[1]

The data's high-priority sectors are its Upgrade sectors, upstream oil and midstream coal. In upstream oil the score is 0.40, and the reserves and trade categories pull it down. No oil reserves are recorded, which the index counts as zero, and the trade deficit in these goods was 1.4% of GDP in 2023. For context, of its USD 5.05 bn of imports of these goods in 2024, Kazakhstan supplied 29.4%, Iraq 14.0% and Saudi Arabia 10.7%. In midstream coal, trade pulls the score down: the deficit was <0.1% of GDP in 2023, a measure on which it scores higher than only 18% of the other economies. For context, Poland supplied 49.5% of its imports of these goods.[1,2,3]

In upstream gas, the same categories pull the score down: no gas reserves are recorded, and the trade deficit was 1.5% of GDP in 2023. For context, its largest suppliers were Azerbaijan (28.6%), Turkiye (16.8%) and Russia (15.4%). In downstream wind, investment and energy access pull the score down. On annual investment it scores higher than only 23% of the other economies, and growth in investment is not reported, which the index counts as zero. Wind power use per person changed by -10.4% over the years to 2025.[1,2,3,4]

## 2. Economic opportunities and why

18 of the 29 placed sectors sit in the two high-opportunity corners. Economic Opportunity measures how well placed a country is to create value in a sector. Compared with other economies, it is highest in downstream batteries, upstream electric grid and downstream solar, higher than 96%, 92% and 88% of the other economies scored. All sit in the Grow corner, which the paper's framework associates with export promotion.[1]

In downstream batteries, investment pushes the score up: it scores higher than 95% of the other economies on annual investment and 96% on its growth. In upstream electric grid, trade pushes the score up: its revealed comparative advantage, the weight of these goods in its exports relative to world exports, was 6.79 in 2024. On export feasibility, which measures related capabilities, it scores higher than 89% of the other economies. For context, its largest markets were Germany (30.9%), Italy (23.9%) and Turkiye (18.6%).[1,2,5]

In downstream solar, energy access and investment push the score up. Solar power use was 3.66 GJ per person in 2025, after a change of 380.1% over the years to 2025. On annual investment it scores higher than 80% of the other economies.[1,4]

## 3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by opportunity, security need, policy activity, stage and low-emission value. On the paper's weighting, it ranks upstream batteries 1st, midstream batteries 2nd, upstream solar 3rd, upstream nuclear 4th and upstream wind 5th among Bulgaria's own sectors. The scores are almost tied: upstream batteries leads with 0.66, just under 0.010 ahead of the next, and the later gaps are 0.005, 0.005 and 0.005.[1]

The data's high-priority sectors rank far lower: upstream oil is 24th and midstream coal 26th. In upstream oil, the technology's low-emission term adds 0.000 and the policy term 0.032, against 0.097 and 0.136 in upstream batteries.[1]

The allied pick, the leading upstream or midstream sector after a screen for resource endowment, is midstream batteries (0.65), 0.010 ahead of upstream nuclear. The screen leaves out upstream batteries and upstream solar because no resource base is recorded for them.[6]

Policy activity counts International Energy Agency Policies and Measures records and reads the Climate Action Tracker rating against each technology. Bulgaria carries the rating Insufficient (November 2024) and has 64 such records, 34 of which name a technology. Comparing priority, the index without its policy term, with policy activity, top priority meets the most policy in upstream batteries, midstream batteries, midstream electric grid and midstream solar. No top-priority sector falls among those with the least policy, and no bottom-priority sector among those with the most. Priority and policy agree at the bottom in 7 sectors, every coal and oil sector among them.[1,7,8]

## 4. Partnership priorities and why

The Partnership Strength Index rates partners as reliable suppliers (Friendshore), as export markets (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. From Bulgaria's viewpoint it ranks Italy 1st, Germany 2nd, the Netherlands 3rd, Greece 4th and Romania 5th of 192 partners. Italy, the Netherlands, Greece and Romania each stand out as an export market; for Italy the best

sectors are upstream electric grid, midstream solar, midstream electric grid, midstream batteries and midstream wind. Germany stands out as a reliable supplier.[1]

Today's trade matches most of this list. Germany is Bulgaria's 1st trade partner on average across its energy-goods baskets, Italy its 3rd, Romania its 5th and Greece its 6th; each is a priority partner with strong ties today. The Netherlands trades less, ranking 15th. All of these partners vote in the same bloc as Bulgaria at the UN and share membership of NATO with it. By contrast, China is its 2nd trade partner but ranks 46th on Partnership Strength, and Turkiye is its 4th but ranks 36th. In UN votes, Bulgaria and China sit in opposite blocs, while Turkiye is nonaligned.[1,2,9,10]

As a partner to the coalition, Bulgaria ranks 26th of 192 economies on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads upstream and midstream production across coalition members, its largest designed shares are 0.2% of upstream coal and 0.2% of midstream wind.[1,6]

## 5. Data notes and caveats

Values are as of 12 October 2026; Country Insights has more.

- Scores compare Bulgaria with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- In the allied network design the European Union is one producer; Bulgaria's shares are derived from its weight within the EU.
- The Climate Action Tracker rates the EU as a whole; Bulgaria carries the EU's rating.
- Bulgaria's counts leave out the EU's own 438 International Energy Agency policy records.
- Trade figures count trade with other EU members.

---


Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
2. United Nations, UN Comtrade Database, HS6 bilateral trade (all reporters), 2020 and 2024; reconciled by the authors (pulled 2026-10-07 to 2026-10-09); attribution; https://comtradeplus.un.org/.
3. Growth Lab at Harvard University, The Atlas of Economic Complexity, HS92 6-digit country-product-year data (data to 2023); attribution; https://atlas.hks.harvard.edu/.
4. Energy Institute, Statistical Review of World Energy 2026 (data to 2025); attribution; https://www.energyinst.org/statistical-review.
5. United Nations, UN Comtrade Database, exports to World by reporter, 2024 (2024 data); attribution; https://comtradeplus.un.org/.
6. Allied network design (the paper's model design of coalition production shares).
7. Climate Action Tracker, country ratings (ratings updated 2022-2025); attribution; https://climateactiontracker.org/.
8. International Energy Agency, Policies and Measures Database (export of 2026-09-30); attribution; https://www.iea.org/policies.

9. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.
10. Economic-security agreements: G7, NATO, Indo-Pacific Economic Framework, Minerals Security Partnership (and Forum), U.S. free-trade agreements (compiled by the authors from official membership lists).

---

*AI-generated from the data and checked against it; paper baseline as of 12 October 2026.*
*Allied Industrial Base Explorer, Country Insights: Bulgaria*